ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 6 - Due date 02/27/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A06_Sp26.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: "ggplot2", "forecast", "tseries" and "sarima". Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.

```
#Load/install required package here
library(ggplot2)
library(forecast)
library(tseries)
library(tidyverse)
library(cowplot)
library(sarima)
```

This assignment has general questions about ARIMA Models.

Q1

Describe the important characteristics of the sample autocorrelation function (ACF) plot and the partial sample autocorrelation function (PACF) plot for the following models:

- AR(2)

  Answer: The ACF plot of an AR(2) model will show exponential decay over time. The PACF plot will show a quick drop off and have insignificant values after 2 lags, which is indicative of a 2nd order AR model.

- MA(1)

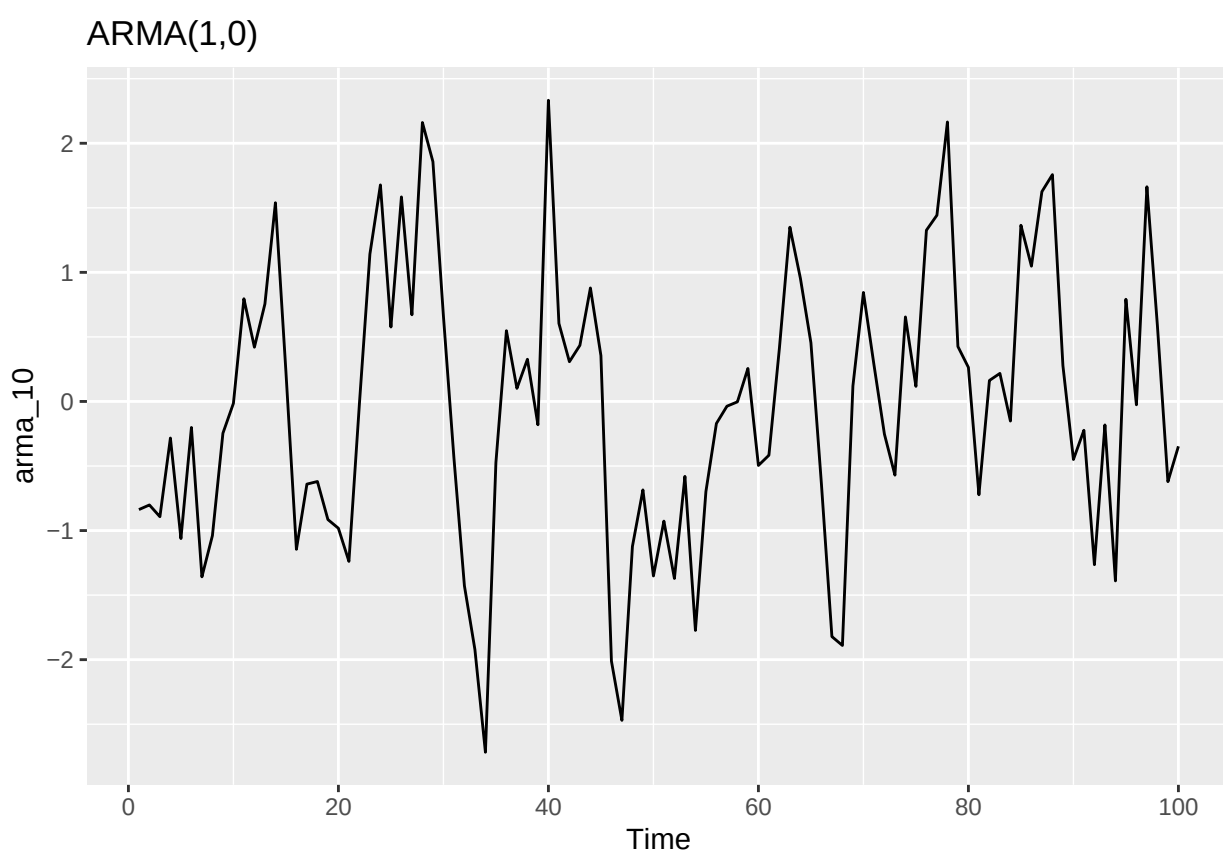
  Answer: The ACF plot of an MA(2) model will have a quick drop off and have insignificant values after 1 lag, which is indicative of the 1st order MA model. The PACF plot will show exponential decay over time.

Q2

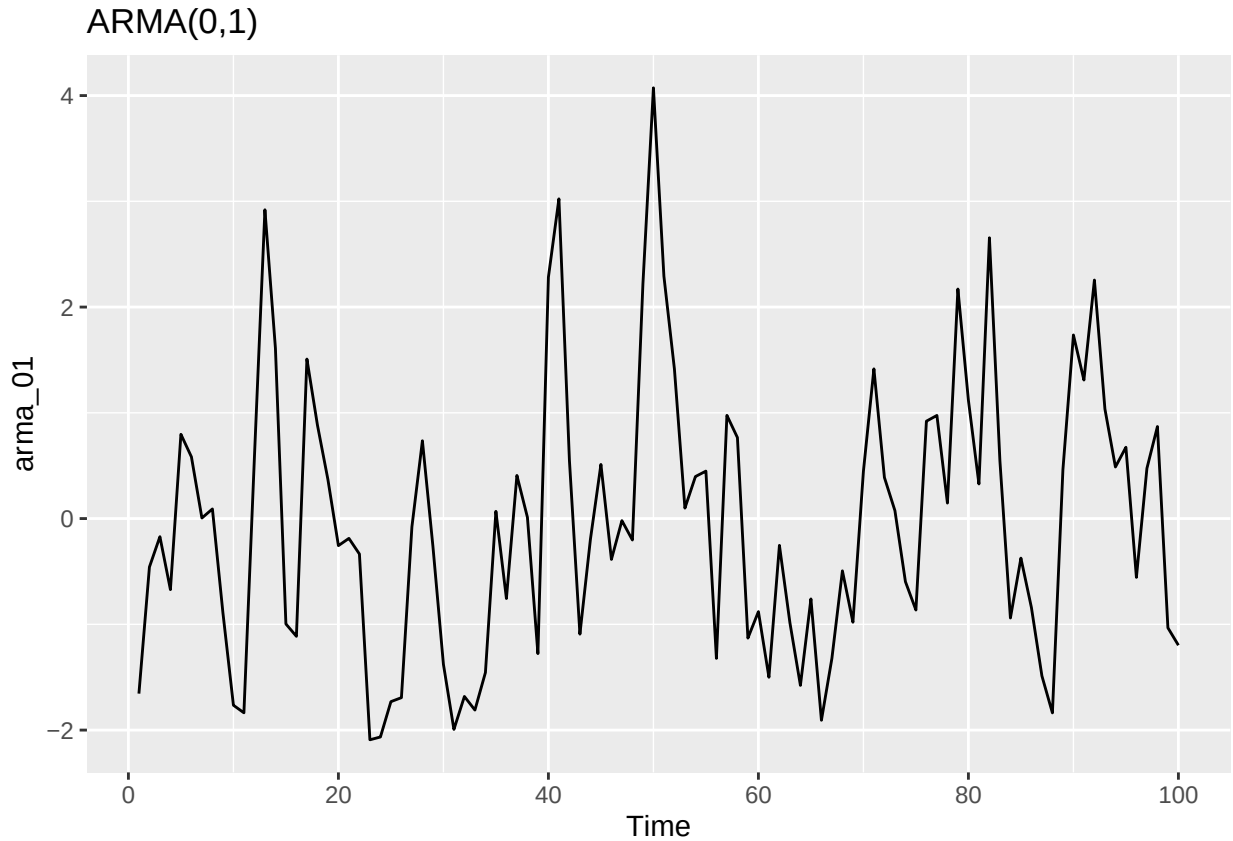
Recall that the non-seasonal ARIMA is described by three parameters ARIMA(p, d, q) where p is the order of the autoregressive component, d is the number of times the series need to be differenced to obtain stationarity and q is the order of the moving average component. If we don't need to difference the series, we don't need to specify the "I" part and we can use the short version, i.e., the ARMA(p, q).

(a) Consider three models: ARMA(1,0), ARMA(0,1) and ARMA(1,1) with parameters $\phi = 0.6$ and $\theta = 0.9$. The ϕ refers to the AR coefficient and the θ refers to the MA coefficient. Use the `arima.sim()` function in R to generate $n = 100$ observations from each of these three models. Then, using `autoplot()` plot the generated series in three separate graphs.

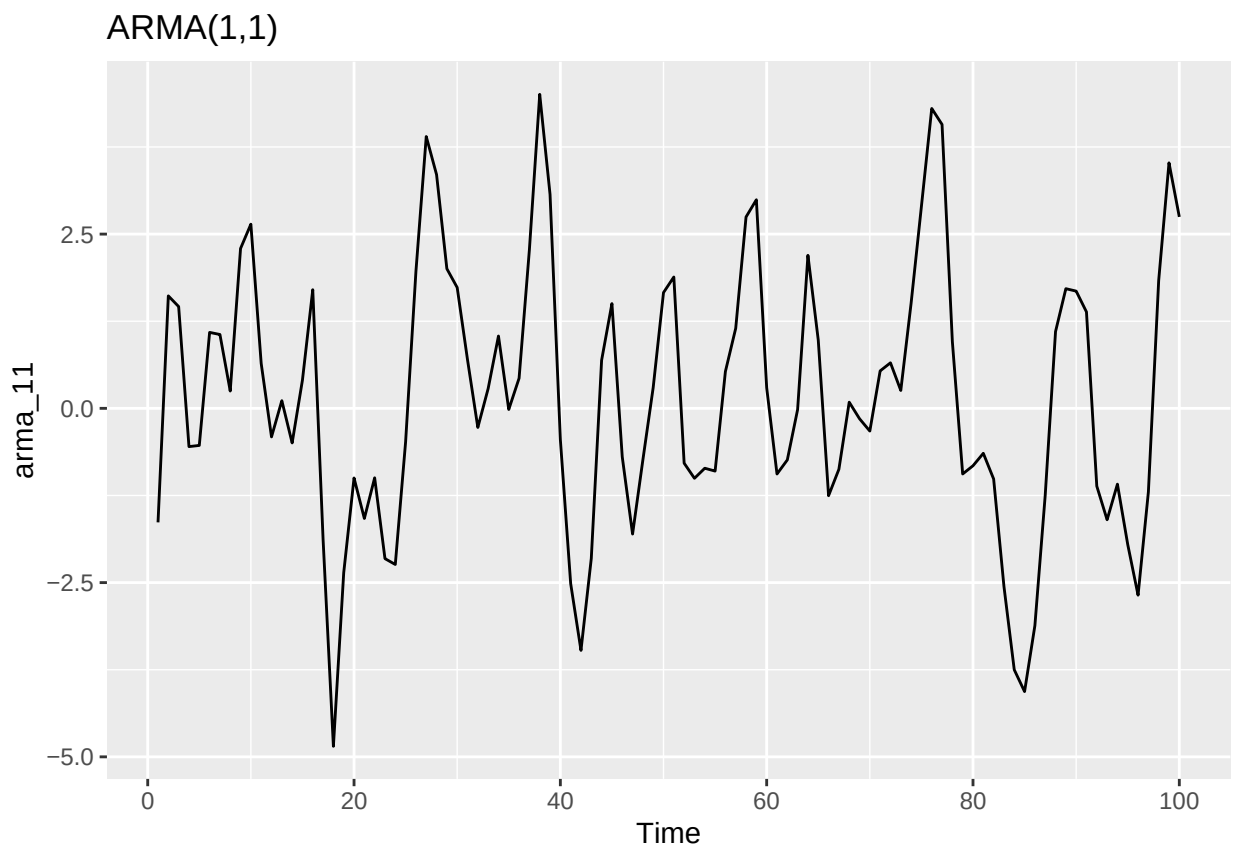
```
set.seed(0)  # set seed so my results are always the same when I knit
arma_10 <- arima.sim(n=100, model=list(order=c(1,0,0), ar=0.6))
autoplot(arma_10) + labs(title = "ARMA(1,0)")
```



```r
arma_01 <- arima.sim(n=100, model=list(order=c(0,0,1), ma=0.9))
autoplot(arma_01) + labs(title = "ARMA(0,1)")
```

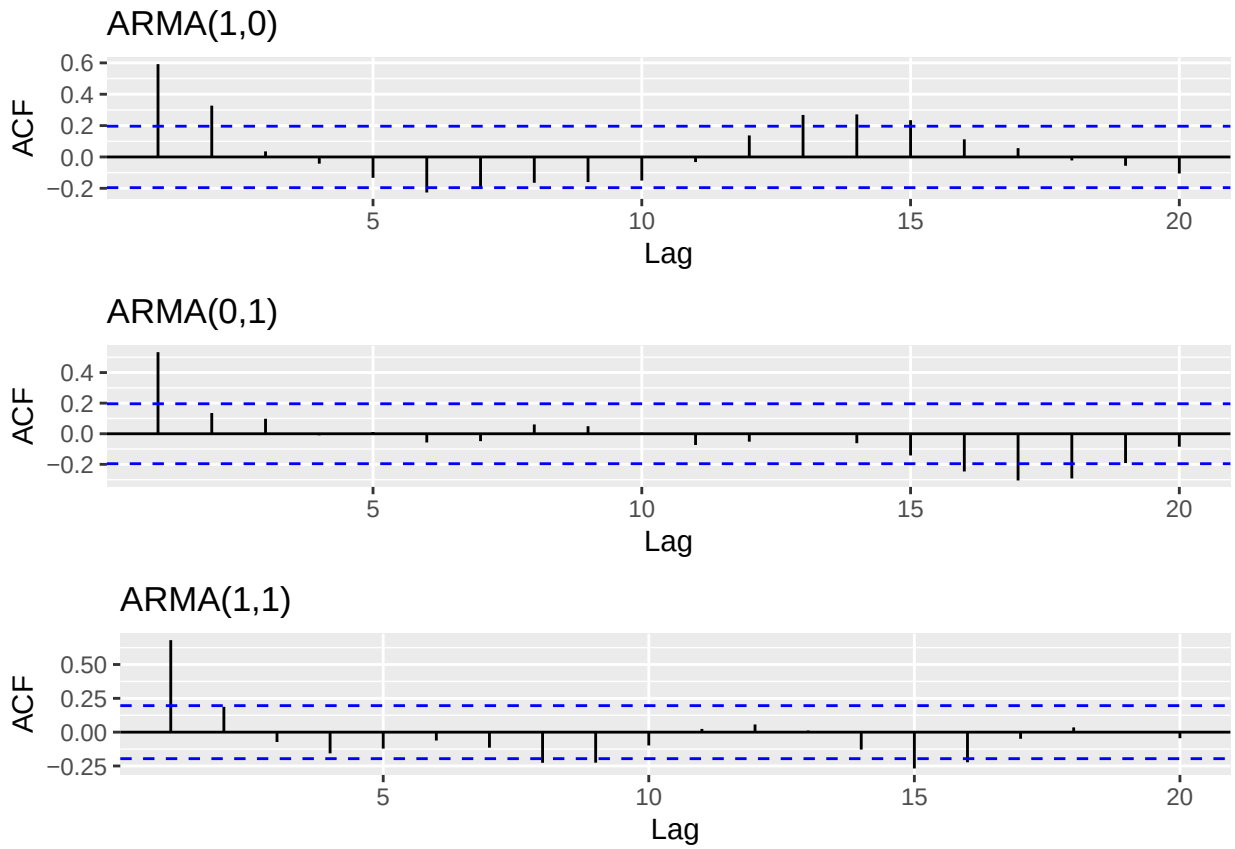



```r
arma_11 <- arima.sim(n=100, model=list(order=c(1,0,1), ar=0.6, ma=0.9))
autoplot(arma_11) + labs(title = "ARMA(1,1)")
```



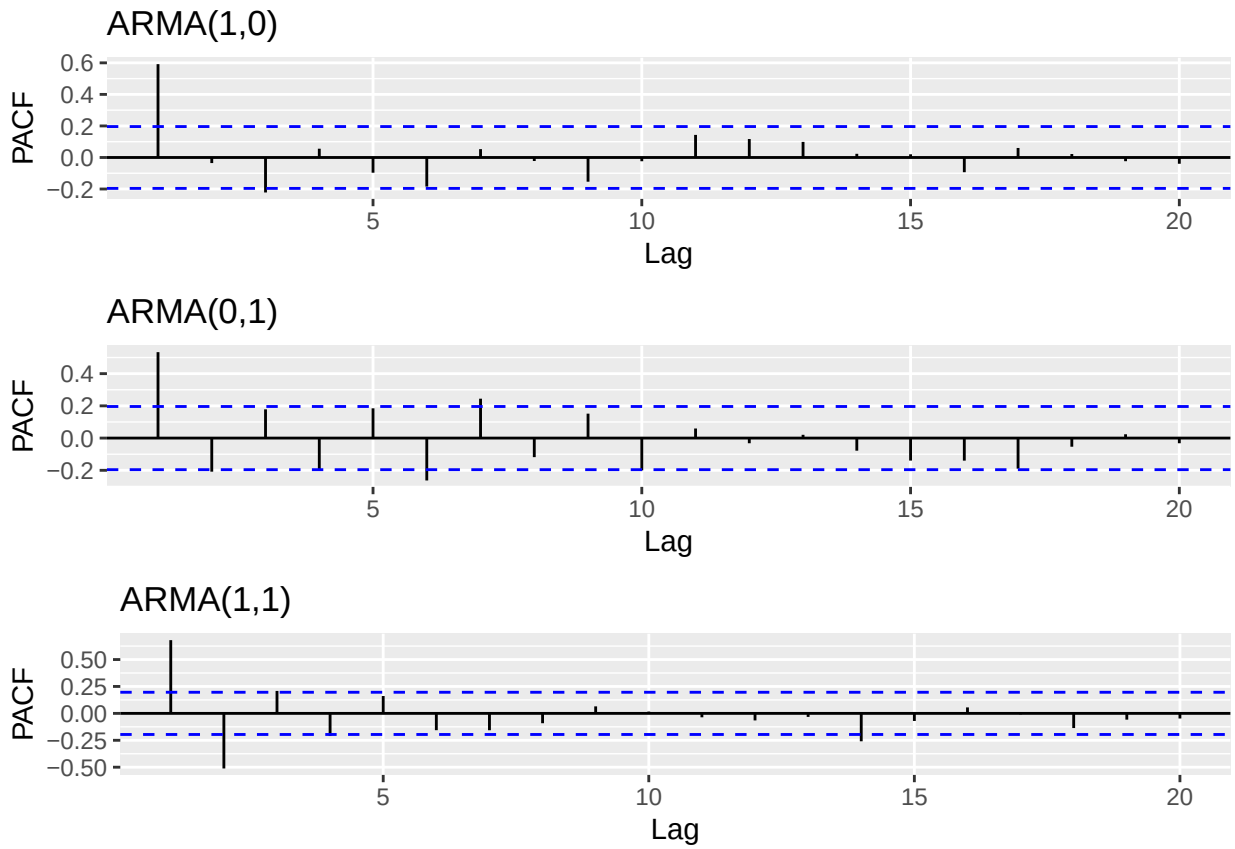
(b) Plot the sample ACF for each of these models in one window to facilitate comparison (Hint: use `cowplot::plot_grid()`).

```
plot_grid(
  ggAcf(arma_10) + labs(title="ARMA(1,0)"),
  ggAcf(arma_01) + labs(title="ARMA(0,1)"),
  ggAcf(arma_11) + labs(title="ARMA(1,1)"),
  nrow = 3
)
```



(c) Plot the sample PACF for each of these models in one window to facilitate comparison.

```
plot_grid(
  ggPacf(arma_10) + labs(title="ARMA(1,0)"),
  ggPacf(arma_01) + labs(title="ARMA(0,1)"),
  ggPacf(arma_11) + labs(title="ARMA(1,1)"),
  nrow = 3
)
```



(d) Look at the ACFs and PACFs. Imagine you had these plots for a data set and you were asked to identify the model, i.e., is it AR, MA or ARMA and the order of each component. Would you be able identify them correctly? Explain your answer.

Answer: It would be difficult to identify these models with such few observations, though the patterns are mostly as I'd expect. For the ARMA(1,0) there is exponential decay of the ACF and a sharp drop off after the first lag of the PACF, however the ACF has negative values after about 5 lags which isn't what I'd expect. For the ARMA(0,1), there is a sharp drop off after one lag of the ACF, which I'd expect. However, the PACF does not appear to have many significant values after the first lag, so it'd be difficult to identify this model with those results. The ARMA(1,1) is also difficult. The ACF drops off after the first term, while the PACF has about two significant lags before droping off. You would not be able to determine the order of this model just looking at the ACF/PACF.

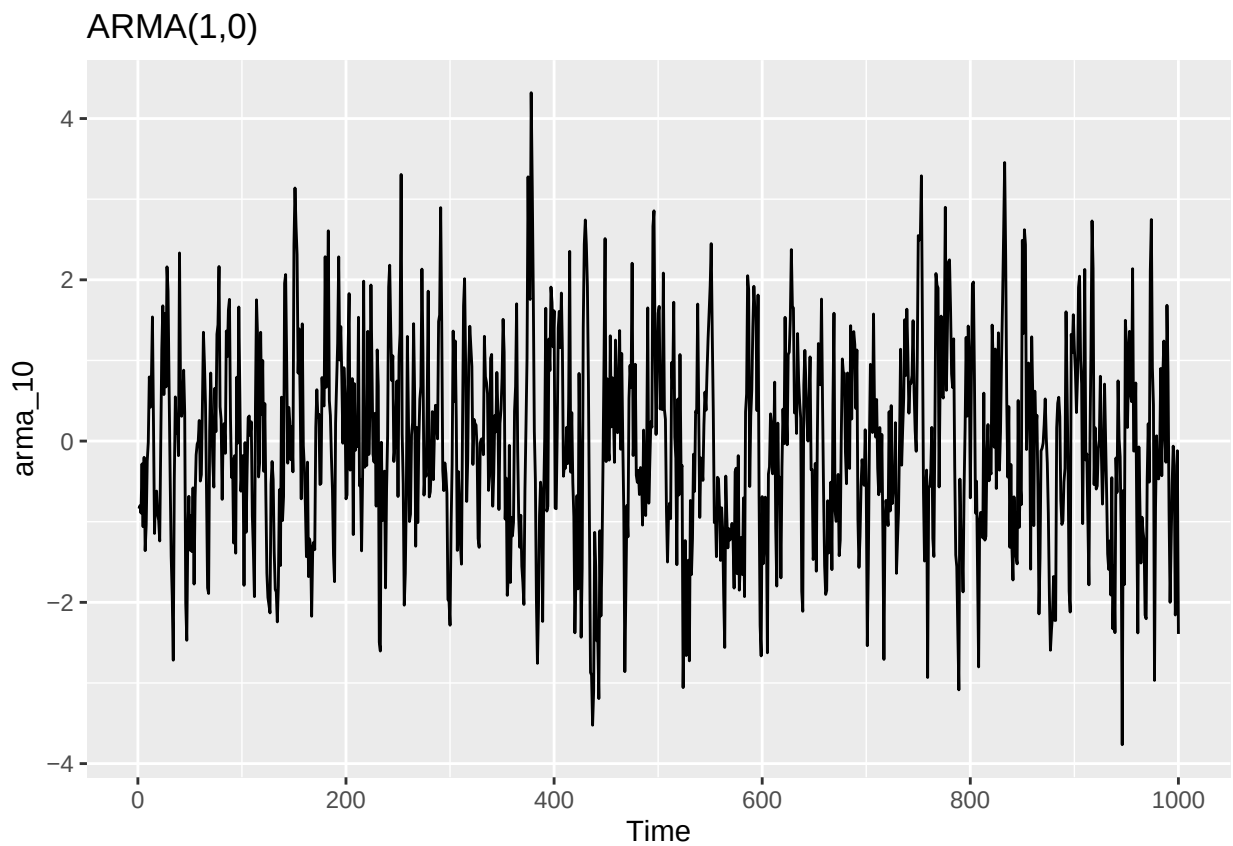
(e) Compare the PACF values R computed with the values you provided for the lag 1 correlation coefficient, i.e., does $\phi = 0.6$ match what you see on PACF for ARMA(1,0), and ARMA(1,1)? Should they match?

Answer: Yes, the value of the PACF of the first lag for both models appears to be around 0.6.

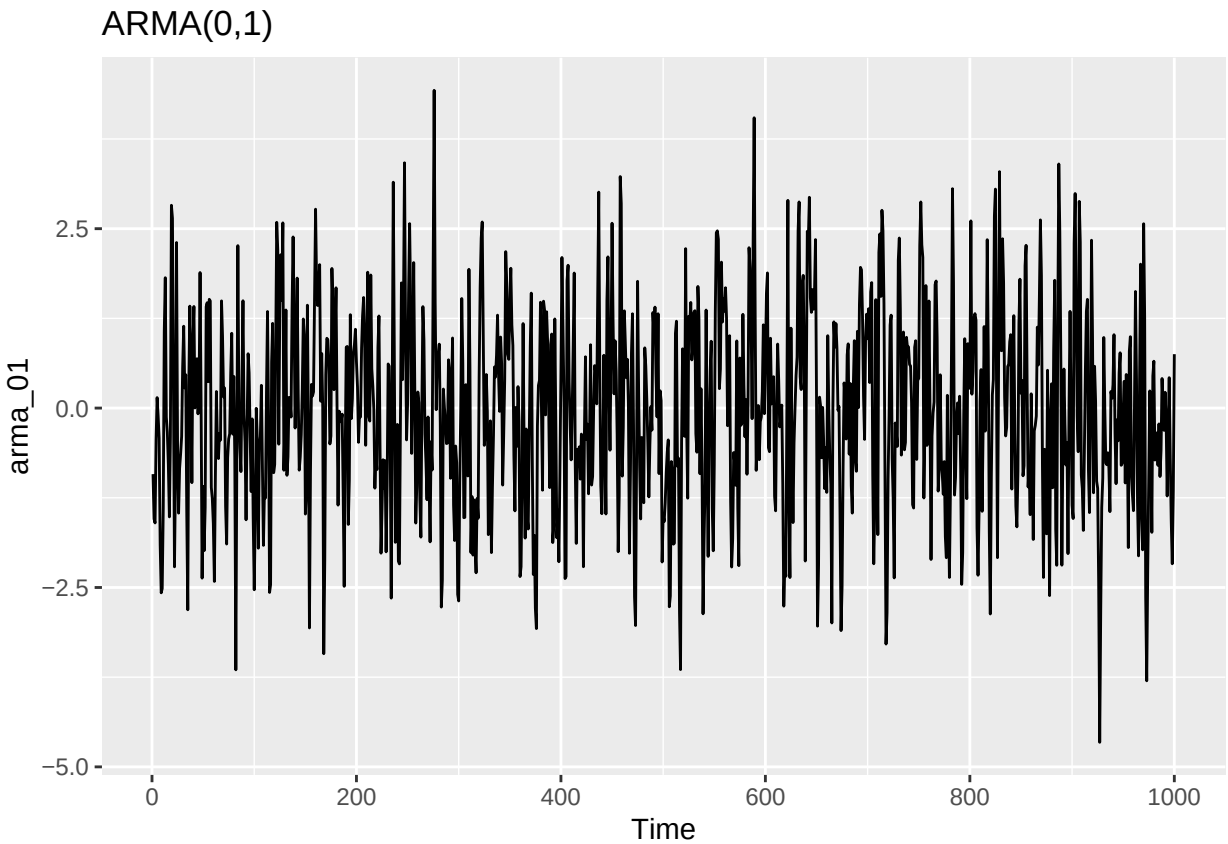
(f) Increase number of observations to $n = 1000$ and repeat parts (b)-(e).

```
set.seed(0)  # set seed for reproducibility
arma_10 <- arima.sim(n=1000, model=list(order=c(1,0,0), ar=0.6))
arma_01 <- arima.sim(n=1000, model=list(order=c(0,0,1), ma=0.9))
```

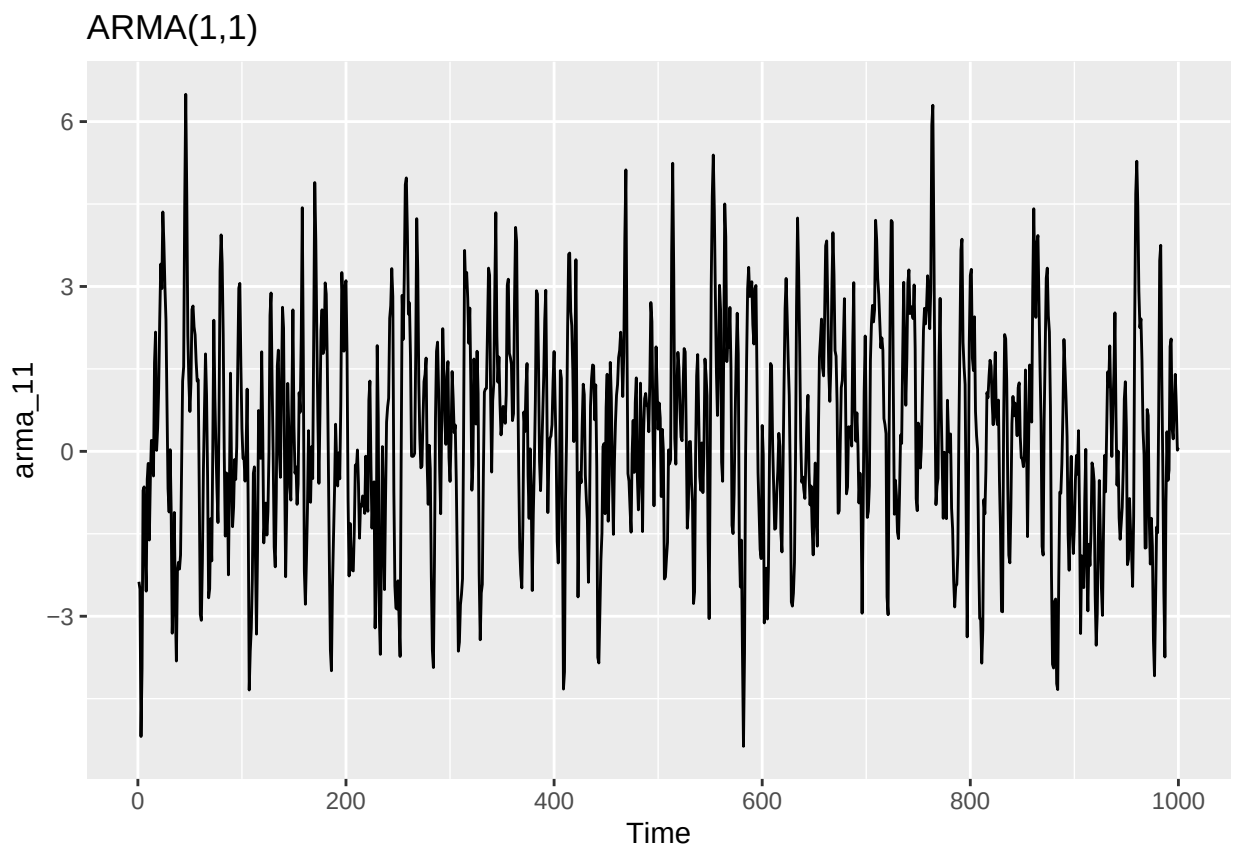
```r
arma_11 <- arima.sim(n=1000, model=list(order=c(1,0,1), ar=0.6, ma=0.9))

autoplot(arma_10) + labs(title = "ARMA(1,0)")
```



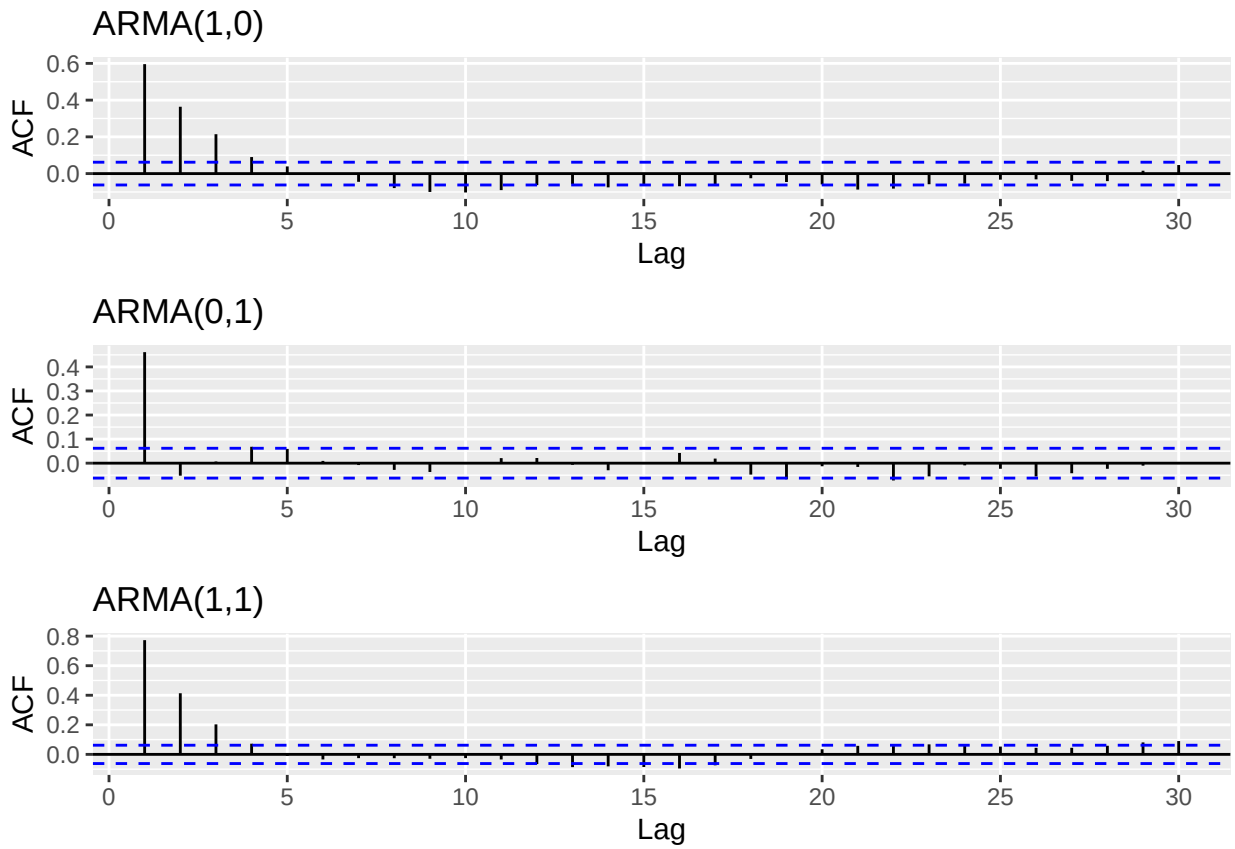

```r
autoplot(arma_01) + labs(title = "ARMA(0,1)")
```



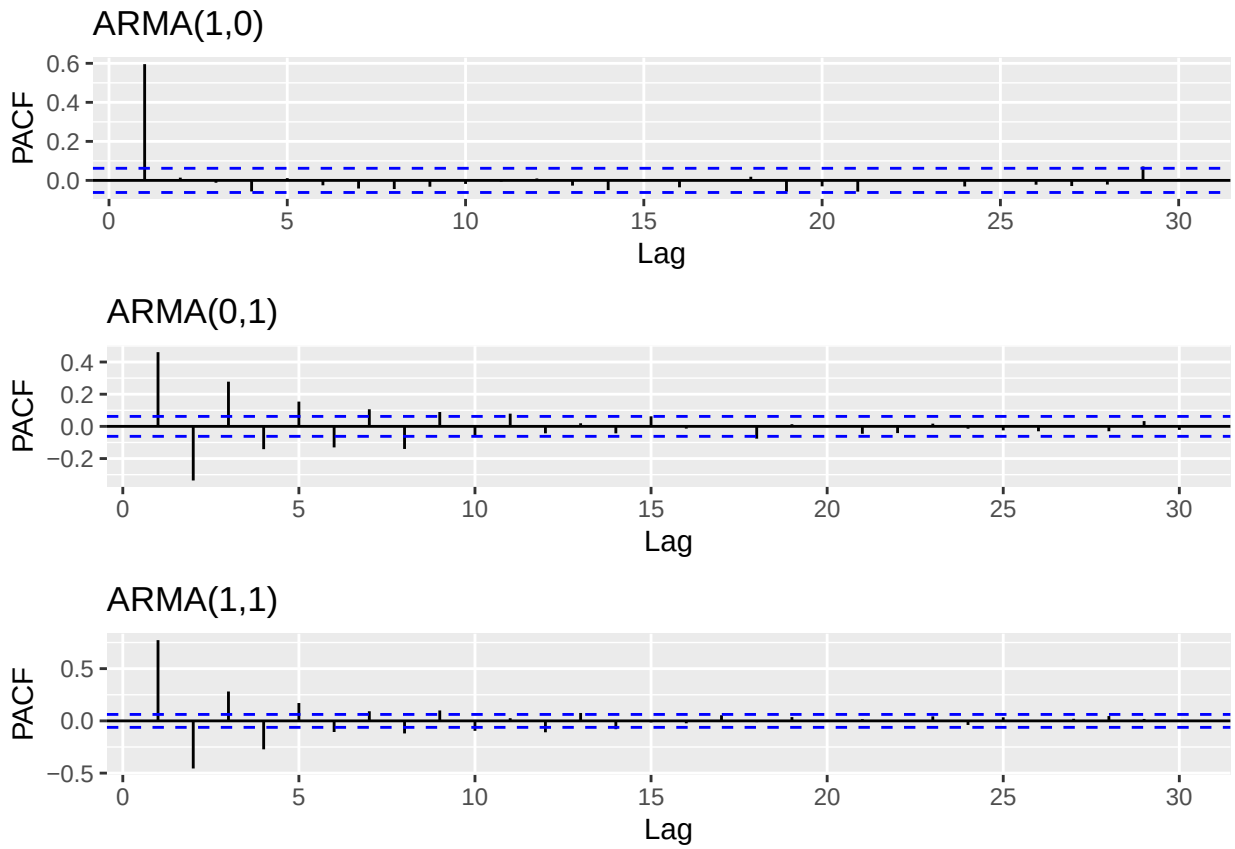

```
autoplot(arma_11) + labs(title = "ARMA(1,1)")
```




```
plot_grid(
  ggAcf(arma_10) + labs(title="ARMA(1,0)"),
  ggAcf(arma_01) + labs(title="ARMA(0,1)"),
  ggAcf(arma_11) + labs(title="ARMA(1,1)"),
  nrow = 3
)
```




```
plot_grid(
  ggPacf(arma_10) + labs(title="ARMA(1,0)"),
  ggPacf(arma_01) + labs(title="ARMA(0,1)"),
  ggPacf(arma_11) + labs(title="ARMA(1,1)"),
  nrow = 3
)
```



Answer: The models are much easier to identify with more observations. For the ARMA(1,0) model, there clear is exponential decay of the ACF, while the PACF drops off completely after one lag. The opposite is true for the ARMA(0,1) model. The PACF is signficiant for the first 7 or so lags, while the ACF drops off completly after the first lag. The ARMA(1,1) is less clear to me. It appears to have exponential decay of both the PACF and ACF, so this suggests both the AR and MA components have an influence on the series. However, you would not be able to determine the order of the components from these results. Again, the value of the first lag of the PACF for both the ARMA(1,0) and ARMA(1,1) models is roughly 0.6.

Q3

Consider the ARIMA model $y_t = 0.7 * y_{t-1} - 0.25 * y_{t-12} + a_t - 0.1 * a_{t-1}$

(a) Identify the model using the notation ARIMA$(p, d, q)(P, D, Q)_s$, i.e., identify the integers p, d, q, P, D, Q, s (if possible) from the equation.

Answer: ARIMA(1, d, 1)(1, D, 0)[12]. You are not able to determine whether the series requires differencing from this equation alone.

(b) Also from the equation what are the values of the parameters, i.e., model coefficients.

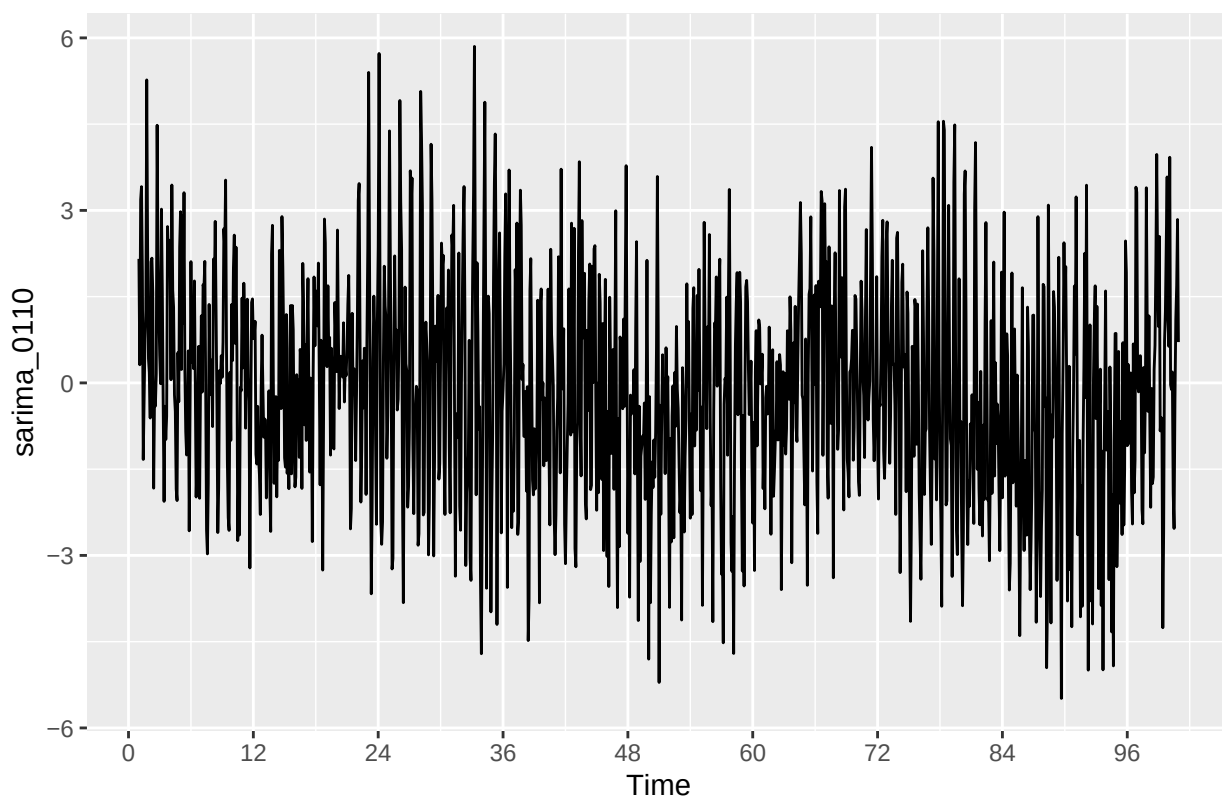
Answer: phi for the first order AR is 0.7, theta for the first order MA is 0.1, and phi for the first order SAR is -0.25.

Q4

Simulate a seasonal ARIMA$(0,1)\times(1,0)_{12}$ model with $\phi = 0.8$ and $\theta = 0.5$ using the `sim_sarima()` function from package `sarima`. The 12 after the bracket tells you that $s = 12$, i.e., the seasonal lag is 12, suggesting monthly data whose behavior is repeated every 12 months. You can generate as many observations as you like. Note the Integrated part was omitted. It means the series do not need differencing, therefore $d = D = 0$. Plot the generated series using `autoplot()`. Does it look seasonal?

```
set.seed(0)  # set seed for reproducibility
sarima_0110 <- sim_sarima(n=1200, model = list(nseasons=12, ma=0.5, sar=0.8)) %>%
  ts(frequency = 12)

autoplot(sarima_0110) +
  scale_x_continuous(breaks=seq(0, 101, by = 12))
```

```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```

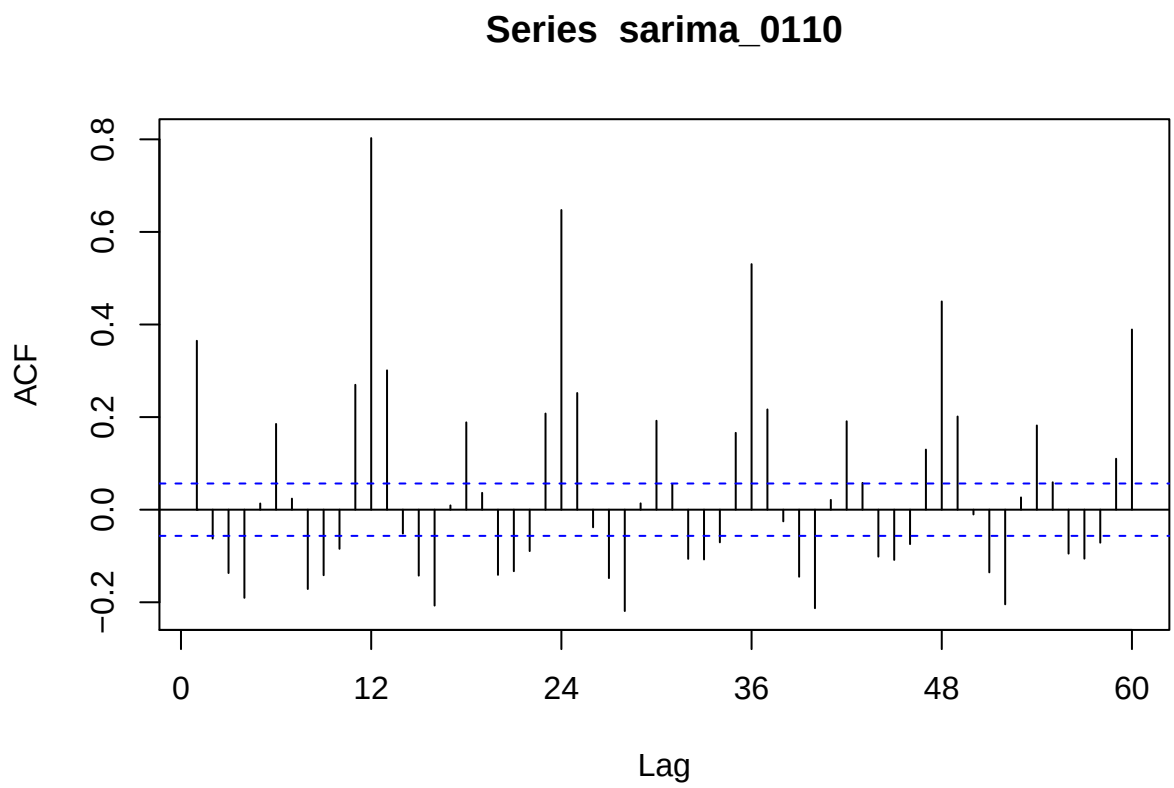


> Answer: There appears to be some seasonal fluctuation but the seasonality is not strong.

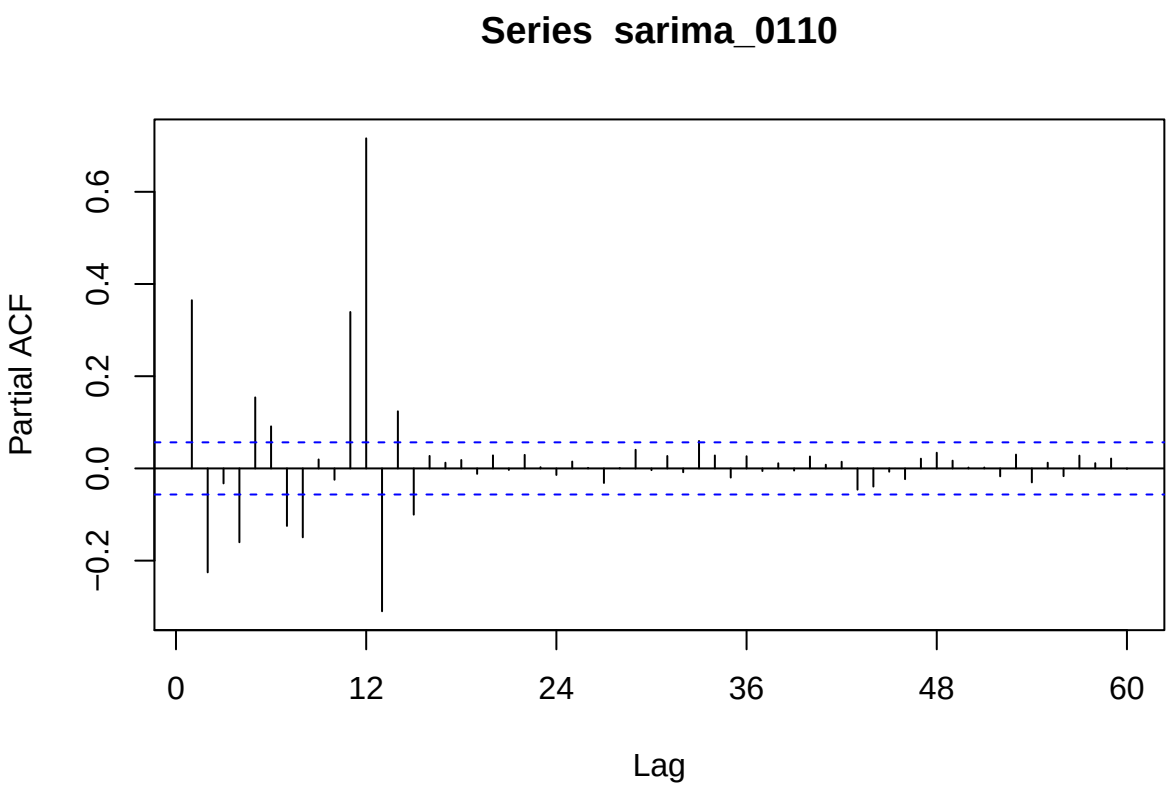
Q5

Plot ACF and PACF of the simulated series in Q4. Comment if the plots are well representing the model you simulated, i.e., would you be able to identify the order of both non-seasonal and seasonal components from the plots? Explain.

```r
Acf(sarima_0110, lag.max=60)
```




```r
Pacf(sarima_0110, lag.max=60)
```



Answer: I believe I would be able to identify the seasonal components from the plots. The ACF has clear seasonality and an exponential decay of seasonal lags, which is consistent with the seasonal AR component. The PACF has one strong spike at lag 12, which confirms that the order of the SAR is 1. It would be more difficult to identify the non-seasonal components from the plots. I may be able to infer that AR=0, as the ACF is not consistent with a strong AR component because most lags are insignificant with the exception of the seasonal lags. I would have a hard time recognizing the first order MA component, although you do see initially negative values in the ACF and a slow decay of PACF values, which are both consistent with MA. The small spike at lag 1 of the ACF may confirm the MA order of 1.